Jaeyoung Cha

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Personal Profiles

I am a second-year Ph.D. student in the Optimization & Machine Learning (OptiML) Laboratory at Kim Jaechul Graduate School of AI, Korea Advanced Institute of Science and Technology (KAIST AI), where I am fortunate to be advised by Prof. Chulhee Yun. Previously, I completed my M.Sc. in AI at KAIST and B.E. in Cyber Defense (minor in Mathematics) at Korea University.

Research Interests

I study the theoretical foundations of modern deep learning and design algorithms grounded in rigorous analysis. My work [C1, C4] establishes lower bounds on convergence rates for without-replacement SGD. Another line of work [C2, C3] develops and theoretically justifies an algorithm that strengthens length generalization in Transformers. Current projects investigate speculative decoding, where I am working on designing sampling strategies to improve inference efficiency given a tree-structured draft. I am also studying formal constructions showing how Transformers can perform in-context inference in hidden Markov models.

Education

Korea Advanced Institute of Science and Technology — *Seoul, South Korea*
Ph.D. in Artificial Intelligence — *Mar. 2024 – present*
- Advisor: Prof. Chulhee Yun

Korea Advanced Institute of Science and Technology — *Seoul, South Korea*
M.Sc. in Artificial Intelligence — *Mar. 2022 – Feb. 2024*
- Advisor: Prof. Chulhee Yun
- Thesis: *Optimal Convergence Rate of Without-Replacement SGD*
- GPA: 4.23/4.3

Korea University — *Seoul, South Korea*
B.E. in Cyber Defense — *Mar. 2018 – Feb. 2022*
- Minor in Mathematics
- Overall GPA: 4.38/4.5 (Ranked 1st/30), Minor GPA: 4.5/4.5

Seoul Science High School — *Seoul, South Korea*
High School — *Mar. 2015 – Feb. 2018*
- GPA: 4.15/4.3

Publications

Published

*: Equal Contribution.

[C4] Yujun Kim*, **Jaeyoung Cha***, Chulhee Yun. Incremental Gradient Descent with Small Epoch Counts is Surprisingly Slow on Ill-Conditioned Problems. **ICML 2025**. Openreview

[C3] Hanseul Cho*, **Jaeyoung Cha***, Srinadh Bhojanapalli, Chulhee Yun. Arithmetic Transformers Can Length-Generalize in Both Operand Length and Count. **ICLR 2025**. Openreview

[C2] Hanseul Cho*, **Jaeyoung Cha***, Pranjal Awasthi, Srinadh Bhojanapalli, Anupam Gupta, Chulhee Yun. Position Coupling: Improving Length Generalization of Arithmetic Transformers Using Task Structure. **Neurips 2024**. Openreview
- Short version in ICML 2024 Workshop on Long-Context Foundation Models (LCFM).

[C1] **Jaeyoung Cha**, Jaewook Lee, Chulhee Yun. Tighter Lower bounds for Shuffling SGD: Random Permutations and Beyond. **ICML 2023**. Openreview
- **Oral** (Top 155/6,538=2.37%)

Awards

2024 **Top Reviewer**, NeurIPS 2024

2023 **Samsung Electronics DS Division Outstanding Collaboration Paper Award**, Samsung

2023 **Oral Presentation**, ICML 2023

2020 - 2021 **National Science & Technology Scholarship**, Korea Student Aid Foundation

2018 - 2021 **Military Scholarship**, Ministry of National Defense, Republic of Korea

Talks

Aug 2024. **SNU-KAIST AI/ML Theory Workshop, Gangneung, South Korea**. *Tight Bounds and Convergence Analysis of Shuffling SGD for Finite-Sum Minimization.*

Jun 2023. **KAIST SAARC Workshop, Busan, South Korea**. *Tighter Lower Bounds for Shuffling SGD: Random Permutations and Beyond.*

Teaching Experience

Fall 25 **AI505 Optimization for AI**, Teaching Assistant
Spring 25 **AI709 Advanced Deep Learning Theory**, Teaching Assistant

Academic Services

Conference Reviewer. ICLR (2025–2026), ICML (2025), NeurIPS (2023–2025)

Journal Reviewer. TMLR (2025), JOTA (2025)

Skills

Languages. Korean (*native*), English (*proficient*)

Computer Languages. Python (*including PyTorch*), LaTeX